Technical proof of the extension to higher-dimensional Gaussian channels

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2026-07-19

Abstract. We prove that, in every fixed dimension $d \geq 2$, the capacity-achieving input distribution for the amplitude-constrained vector Gaussian channel must use at least order $A\sqrt{\log A}$ concentric spherical shells. The proof extends the scalar support-size method by combining near-optimality of the uniform input, an L^2 approximation lower bound, and a bridge from relative entropy to L^2 distance. Its main new ingredient is a Slepian-mode argument that separates radial and angular degrees of freedom on the ball.

Technical report HW-TR-2026-01, 19 July 2026. Download the PDF · Download BibTeX

Main result

Consider the amplitude-constrained vector Gaussian channel

$$Y = X + Z, \qquad Z \sim \phi_d = N(0, I_d), \qquad \|X\| \leq A,$$

where the dimension $d \geq 2$ is fixed. Write

$$B_A = \{x \in \mathbb{R}^d : \|x\| \leq A\}$$

for the spatial input ball, and let μ_A be the uniform probability law on B_A. For $0 \leq r \leq A$, let s_r be the uniform probability law on the sphere $\{x : \|x\| = r\}$.

The capacity-achieving input distribution (CAID) is radially symmetric and is supported on finitely many concentric spheres. Thus it has the form

$$\rho_A^* = \sum_{j=1}^{K_A} w_j s_{r_j}, \qquad w_j > 0, \qquad \sum_{j=1}^{K_A} w_j = 1.$$

This structure, as well as the upper bound $K_A = O_d(A^2)$, is due to Dytso, Yagli, Poor, and Shamai. In one dimension, K_A counts atoms, whereas here it counts distinct spherical shells.

For any input law ρ on B_A, let p_ρ denote the Lebesgue density of the output law $\rho * \phi_d$:

$$p_\rho(y) = \int_{B_A} \phi_d(y - x)\, d\rho(x), \qquad \phi_d(y) = \frac{1}{(2\pi)^{d/2}} \exp(-\|y\|^2/2)$$

The main result of this note is the following fixed-dimensional asymptotic lower bound, extending our scalar AWGN results in Wang and Wang, Barletta, and Dytso.

Theorem (shell-count lower bound). For each fixed $d \geq 2$, and sufficiently large A

$$\boxed{K_A \gtrsim_d A\sqrt{\log A}} \qquad \text{(Main result)}$$

The notation $f \gtrsim_d g$ means that $f \geq c_d g$ for some positive constant depending only on d.

Proof of main result

Outline of the proof

The proof and theory construction follow the same architecture as in the scalar channel, with the following correspondence:

	Scalar channel	Vector channel
Reference input	Uniform on $[-A, A]$	Uniform on B_A
CAID complexity	K_A atoms	K_A shells
Spectral analysis	Ordinary Fourier modes	Slepian modes separated into spherical harmonics
Approximation metric	χ^2 of the outputs	L^2 norm of outputs' pdf

The proof for the vector channel follows the same three steps as in the scalar case.

Step 1: Jeffreys is near the CAID

The uniform law μ_A is the vector analogue of the scalar Jeffreys reference input. Its output is close to the optimal output in reverse KL divergence:

$$\boxed{D(p_{\mu_A} \| p_{\rho^*_A}) \lesssim_d \frac{1}{A}} \qquad \text{(Jeffreys–CAID)}$$

Step 2: Approximation lower bound

If ρ is supported on only K concentric shells, its output cannot approximate the Jeffreys output p_{μ_A} too well:

$$\boxed{A^d \|p_\rho - p_{\mu_A}\|_2^2 \gtrsim_d \exp\left(-c_d \frac{K^2}{A^2}\right)}, \qquad \text{(Approx. lower bound)}$$

where the constant c_d depends only on d.

Step 3: Bridging Steps 1 and 2

For the two outputs of interest, reverse KL divergence controls their ordinary L^2 separation:

$$\boxed{D(p_{\mu_A} \| p_{\rho^*_A}) \gtrsim_d A^d \|p_{\rho^*_A} - p_{\mu_A}\|_2^2.} \tag{Bridge}$$

Sketch of the proof of the main result

Applying (Eq. Approx. lower bound), (Eq. Bridge), and (Eq. Jeffreys–CAID) gives

$$\exp\left(-c_d \frac{K_A^2}{A^2}\right) \quad \lesssim_d A^d \|p_{\rho^*_A} - p_{\mu_A}\|_2^2 \quad \lesssim_d D(p_{\mu_A} \| p_{\rho^*_A}) \quad \lesssim_d \frac{1}{A}$$

Taking logarithms yields

$$c_d \frac{K_A^2}{A^2} \geq \log A - O_d(1).$$

For all sufficiently large A, this implies

$$K_A \gtrsim_d A\sqrt{\log A},$$

which is (Eq. Main result).

Step 2, the approximation theory lower bound, is the key difficulty. In the rest of the section, we prove it first, then we prove the more standard results of Steps 1 and 3.

Approximation lower bound

Precise statement and conventions

For simplicity, fix $d \geq 2$, take A to be sufficiently large, and let

$$\rho = \sum_{j=1}^{K} w_j s_{r_j}, \qquad 0 \leq r_j \leq A,$$

be a probability law supported on K concentric shells. Throughout this section, $\lesssim$, $\gtrsim$, and $\asymp$ hide positive prefactors that depend only on d; we usually suppress the subscript d.

The goal is to prove (Eq. Approx. lower bound). The argument has three components:

1. construct about Ω^d bandlimited modes concentrated on the ball B_A;
2. show that one shell can populate only about Ω^{d-1} independent angular channels;
3. transfer the resulting rank defect through Gaussian convolution at cost of factor $\exp(c_d\Omega^2/A^2)$.

Choosing $\Omega \asymp K$ then gives the desired exponent.

Fourier setup and the projection kernel

We use the unitary Fourier transform

$$\hat{f}(\xi) = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-i\xi\cdot x} f(x)\, dx.$$

The spectral (frequency) radius Ω is a positive number, and we consider the space of bandlimited frequencies:

$$B_\Omega = \{\xi \in \mathbb{R}^d : \|\xi\| \le \Omega\}$$

and define the corresponding space of bandlimited functions

$$PW_\Omega = \{f \in L^2(\mathbb{R}^d) : \hat{f}(\xi) = 0 \text{ for } \xi \notin B_\Omega\}.$$

Thus PW_Ω is simply the space of functions with no frequencies larger than Ω. Let $P_\Omega : L^2(\mathbb{R}^d) \to PW_\Omega$ be the projection:

$$\widehat{P_\Omega f} = 1_{B_\Omega} \hat{f}.$$

Fourier inversion shows that P_Ω is convolution with the kernel

$$\mathcal{K}_\Omega(z) = \frac{1}{(2\pi)^d} \int_{B_\Omega} e^{i\xi\cdot z}\, d\xi,$$

so that

$$(P_\Omega f)(x) = \int_{\mathbb{R}^d} \mathcal{K}_\Omega(x-y) f(y)\, dy.$$

Lemma (the Fourier–Bessel kernel). Let $V_d = \pi^{d/2}/\Gamma(d/2+1)$ be the volume of the unit ball. For $z \neq 0$,

$$\mathcal{K}_\Omega(z) = (2\pi)^{-d/2} \Omega^d \frac{J_{d/2}(\Omega\|z\|)}{(\Omega\|z\|)^{d/2}},$$

while

$$\mathcal{K}_\Omega(0) = \frac{V_d}{(2\pi)^d} \Omega^d.$$

Moreover, we have the estimate

$$|\mathcal{K}_\Omega(z)| \lesssim \Omega^d (1 + \Omega\|z\|)^{-(d+1)/2}. \qquad \text{(Kernel decay)}$$

Proof. The displayed identity is the standard radial Fourier-transform formula for the indicator of a ball. For fixed order $\nu \ge 0$, the Bessel function satisfies

$$|J_\nu(t)| \le \begin{cases} \dfrac{(t/2)^\nu}{\Gamma(\nu+1)}, & 0 < t \le 1, \\ C_\nu t^{-1/2}, & t \ge 1, \end{cases}$$

where $C_\nu > 0$ is a constant depending on ν. The first line is DLMF 10.14.4; the second follows from the fixed-order large-argument expansion DLMF 10.17.3 together with its real-argument error bound. Equivalently,

$$|J_\nu(t)| \lesssim_\nu \frac{t^\nu}{(1+t)^{\nu+1/2}}, \qquad t > 0.$$

Substituting $\nu = d/2$ and $t = \Omega\|z\|$ into the kernel formula proves (Eq. Kernel decay). $\square$

Slepian modes and the volume-order count

Let $M_{B_1} : L^2(\mathbb{R}^d) \to L^2(\mathbb{R}^d)$ denote multiplication by the indicator 1_{B_1}. Define the concentration operator on $L^2(\mathbb{R}^d)$ by

$$\mathcal{C}_\Omega = P_\Omega \circ M_{B_1} \circ P_\Omega.$$

Both P_Ω and M_{B_1} are self-adjoint, so $\mathcal{C}_\Omega$ is self-adjoint. Its range lies in PW_Ω, and its restriction to PW_Ω is the integral operator

$$(\mathcal{C}_\Omega f)(x) = \int_{B_1} \mathcal{K}_\Omega(x-y) f(y)\, dy.$$

On PW_Ω, this is a compact self-adjoint operator. Its nonzero spectrum consists of a countable sequence

$$1 > \lambda_1 \geq \lambda_2 \geq \cdots > 0, \qquad \lambda_\alpha \longrightarrow 0,$$

with globally orthonormal eigenfunctions $\{\psi_\alpha\}_{\alpha \in \mathbb{N}} \subset PW_\Omega$ satisfying

$$\mathcal{C}_\Omega \psi_\alpha = \lambda_\alpha \psi_\alpha, \qquad \int_{\mathbb{R}^d} \psi_\alpha(x) \overline{\psi_\beta(x)}\, dx = \delta_{\alpha\beta}.$$

These are the Slepian modes. This standard spectral description goes back to the time- and frequency-limiting theory of Landau and Widom; see also the ball-based exposition of Khalid, Kennedy, and McEwen.

The rank argument below needs many directions on which $\mathcal{C}_\Omega$ acts non-negligibly. We therefore count its dominant modes.

Definition (dominant Slepian modes). Define dominant modes of $\mathcal{C}_\Omega$ to be:

$$\mathcal{I}_\Omega = \{\alpha \in \mathbb{N} : \lambda_\alpha \geq 1/2\}.$$

$N_\Omega = |\mathcal{I}_\Omega|$ is the number of dominant Slepian modes.

We now count these modes.

Lemma 1 (volume-order dominant-mode count). For sufficiently large Ω,

$$N_\Omega \asymp \Omega^d. \qquad \text{(Mode count)}$$

Proof.

Firstly, recall the following useful fact. If an integral operator T has square-integrable kernel $t(x,y)$, then

$$\|T\|_{\mathrm{HS}}^2 = \iint |t(x,y)|^2\,dx\,dy, \qquad \mathrm{Tr}(T^*T) = \|T\|_{\mathrm{HS}}^2.$$

Set $R_\Omega = M_{B_1}P_\Omega$, which has the kernel $r_\Omega(x,y) = 1_{B_1}(x)\mathcal{K}_\Omega(x-y)$, then $\mathcal{C}_\Omega = R_\Omega^* R_\Omega$. Therefore,

$$\begin{aligned}
\sum_\alpha \lambda_\alpha &= \mathrm{Tr}(\mathcal{C}_\Omega) = \mathrm{Tr}(R_\Omega^* R_\Omega) \\
&= \|R_\Omega\|_{\mathrm{HS}}^2 \\
&= \int_{B_1}\int_{\mathbb{R}^d} |\mathcal{K}_\Omega(x-y)|^2\,dy\,dx \\
&= \int_{B_1} (2\pi)^{-d}|B_\Omega|\,dx = \frac{V_d^2}{(2\pi)^d}\Omega^d.
\end{aligned}$$

where we used Plancherel to calculate $\|\mathcal{K}_\Omega\|^2$.

Secondly,

$$\begin{aligned}
\sum_{\alpha\geq 1} \lambda_\alpha^2 &= \mathrm{Tr}(\mathcal{C}_\Omega\mathcal{C}_\Omega) = \mathrm{Tr}(R_\Omega^* R_\Omega R_\Omega^* R_\Omega) = \mathrm{Tr}\big(R_\Omega R_\Omega^* R_\Omega R_\Omega^*\big) \\
&= \|R_\Omega R_\Omega^*\|_{\mathrm{HS}}^2 \\
&= \int_{B_1}\int_{B_1} |\mathcal{K}_\Omega(x-y)|^2\,dy\,dx.
\end{aligned}$$

Here we used cyclicity of the trace and the fact that $R_\Omega R_\Omega^*$ is the self-adjoint integral operator with kernel

$$1_{B_1}(x)\mathcal{K}_\Omega(x-y)1_{B_1}(y).$$

Thirdly, subtracting the two identities gives

$$\begin{aligned}
\sum_{\alpha\geq 1} \lambda_\alpha(1-\lambda_\alpha) &= \sum_{\alpha\geq 1}\lambda_\alpha - \sum_{\alpha\geq 1}\lambda_\alpha^2 \\
&= \int_{B_1}\int_{B_1^c} |\mathcal{K}_\Omega(x-y)|^2\,dy\,dx
\end{aligned}$$

Applying (Eq. Kernel decay) and writing $z = y - x$, we obtain

$$\sum_{\alpha\geq 1} \lambda_\alpha(1-\lambda_\alpha) \lesssim \Omega^{2d}\int_{\mathbb{R}^d} (1+\Omega\|z\|)^{-(d+1)} m(z)\,dz,$$

where $m(z) = \big|\{x \in B_1 : x + z \notin B_1\}\big|$ is the volume pushed outside B_1 by translation through z, which admits the following elementary upper bound:

$$m(z) \lesssim \min\{1, \|z\|\}.$$

This allows us to derive the upper bound

$$\sum_{\alpha \geq 1} \lambda_\alpha (1 - \lambda_\alpha) \lesssim \Omega^{d-1} \log(2 + \Omega)$$

Lastly, combining all previous estimates, we have

$$\begin{aligned}
\Omega^d &\asymp \sum_\alpha \lambda_\alpha \\
&= \sum_{\text{dominant } \alpha} \lambda_\alpha + \sum_{\text{non-dominant } \alpha} \lambda_\alpha \\
&\leq \sum_{\text{dominant } \alpha} \lambda_\alpha + \sum_{\text{non-dominant } \alpha} 2\lambda_\alpha (1 - \lambda_\alpha) \\
&\lesssim N_\Omega + \Omega^{d-1} \log(2 + \Omega)
\end{aligned}$$

which leads to the dominant mode count $N_\Omega \asymp \Omega^d$.

Spherical harmonics and the surface-order count

The preceding count uses only volume. To see what a single sphere can contribute, we must separate radial and angular behavior.

Every nonzero $x \in \mathbb{R}^d$ can be written as

$$x = r\theta, \qquad r = \|x\|, \qquad \theta = \frac{x}{\|x\|} \in S^{d-1}.$$

Here r is a scalar radius and θ is a direction on the unit sphere. In $d = 2$, θ can be represented by one ordinary angle; in higher dimensions it is a point of S^{d-1}.

A degree-ℓ spherical harmonic is the restriction to S^{d-1} of a homogeneous harmonic polynomial of degree ℓ. Let $\mathcal{H}_{\ell,d}$ be the space of these functions and choose an orthonormal basis:

$$\{Y_{\ell,m} : 1 \leq m \leq h_{\ell,d}\},$$

where the dimension is $h_{\ell,d} = \dim \mathcal{H}_{\ell,d} = \binom{\ell+d-1}{\ell} - \binom{\ell+d-3}{\ell-2}$.

See Frye and Efthimiou more details on general dimension spherical harmonics.

Proposition (separation of the Slepian modes). For every (ℓ, m), the subspace

$$\mathcal{V}_{\ell,m} = \left\{R(r)Y_{\ell,m}(\theta) : R \in L^2\left((0,\infty), r^{d-1}dr\right)\right\} \cap PW_\Omega$$

is preserved by $\mathcal{C}_\Omega$. The restriction of $\mathcal{C}_\Omega$ to this subspace is the same radial operator for every m at fixed ℓ. Consequently, the Slepian eigenfunctions can be chosen in the separated form

$$\psi_{\ell,m,k}(r\theta) = R_{\ell,k}(r)Y_{\ell,m}(\theta),$$

with eigenvalue $\lambda_{\ell,k}$ independent of m.

Proof. Take $f(s\varphi) = R(s)Y_{\ell,m}(\varphi) \in \mathcal{V}_{\ell,m}$, where $s \geq 0$ and $\varphi \in S^{d-1}$. The integral formula for the concentration operator gives

$$(\mathcal{C}_\Omega f)(r\theta) = \int_0^1 \int_{S^{d-1}} \mathcal{K}_\Omega(r\theta - s\varphi)R(s)Y_{\ell,m}(\varphi)s^{d-1}\, d\varphi\, ds.$$

Because $\mathcal{K}_\Omega$ is radial,

$$\mathcal{K}_\Omega(r\theta - s\varphi) = F_{r,s}(\theta \cdot \varphi), \qquad F_{r,s}(t) = \mathcal{K}_\Omega\left(\sqrt{r^2 + s^2 - 2rst}\right).$$

The Funk–Hecke theorem states that integration against any kernel depending only on $\theta \cdot \varphi$ acts diagonally on spherical harmonics. Hence

$$\int_{S^{d-1}} F_{r,s}(\theta \cdot \varphi)Y_{\ell,m}(\varphi)\, d\varphi = a_{\ell,\Omega}(r,s)Y_{\ell,m}(\theta),$$

where the scalar $a_{\ell,\Omega}(r,s)$ depends on ℓ but not on m. Substitution into the concentration integral gives

$$(\mathcal{C}_\Omega f)(r\theta) = Y_{\ell,m}(\theta) \int_0^1 a_{\ell,\Omega}(r,s)R(s)s^{d-1}\, ds.$$

Thus $\mathcal{C}_\Omega$ preserves $\mathcal{V}_{\ell,m}$ and acts there through the radial integral operator

$$(T_{\ell,\Omega}R)(r) = \int_0^1 a_{\ell,\Omega}(r,s)R(s)s^{d-1}\, ds,$$

which is independent of m. Because $\mathcal{C}_\Omega$ is self-adjoint and $\mathcal{V}_{\ell,m}$ is invariant, its orthogonal complement is invariant as well; hence the restriction is self-adjoint. It is also compact, so the spectral theorem supplies radial eigenfunctions $R_{\ell,k}$ of $T_{\ell,\Omega}$. Finally, the spherical harmonics form a complete orthonormal basis of $L^2(S^{d-1})$, so the eigenbases of these radial sectors combine into a separated Slepian eigenbasis. The three-dimensional version of this Fourier–Bessel separation for Slepian concentration on a ball is also developed by Khalid, Kennedy, and McEwen. □

The proposition asserts that a separated eigenbasis can be chosen. An arbitrary eigenfunction obtained by mixing separated modes with the same eigenvalue need not itself be separated. The indices have distinct roles: ℓ is the angular degree, m selects one angular basis function at that degree, and k enumerates radial eigenfunctions within the same angular channel.

Let

$$\mathcal{A}_\Omega = \{(\ell,m) : \lambda_{\ell,k} \geq 1/2 \text{ for at least one } k\}, \qquad H_\Omega = |\mathcal{A}_\Omega|.$$

Thus N_Ω counts all retained triples (ℓ, m, k), whereas H_Ω counts each active angular pair (ℓ, m) only once.

Lemma 2 (surface-order angular count). For all sufficiently large Ω,

$$H_\Omega \lesssim \Omega^{d-1}. \qquad \text{(Angular count)}$$

Proof. Fix (ℓ, m) and write

$$\nu = \frac{d}{2} - 1, \qquad \alpha = \ell + \nu.$$

We first compute the sum of the eigenvalues in this angular sector. The same Funk–Hecke formula used in the preceding proposition gives the Fourier–Bessel identity

$$\widehat{R(r)Y_{\ell,m}}(\theta)(\varpi\omega) = (-i)^\ell Y_{\ell,m}(\omega)\varpi^{-\nu} \int_0^\infty R(r) J_\alpha(r\varpi) r^{\nu+1}\, dr.$$

Consequently, after identifying the (ℓ, m) sector with its radial coefficient, the map that takes a frequency-limited function to its restriction to B_1 has kernel

$$b_{\ell,\Omega}(r, \varpi) = i^\ell 1_{[0,1]}(r) 1_{[0,\Omega]}(\varpi)(r\varpi)^{-\nu} J_\alpha(r\varpi)$$

with respect to the radial measures $r^{d-1}dr$ and $\varpi^{d-1}d\varpi$. As in the trace computation of Lemma 1, the sum of the sector eigenvalues is the squared Hilbert–Schmidt norm of this restriction map. Therefore

$$S_{\ell,\Omega} := \sum_k \lambda_{\ell,k} = \int_0^1 \int_0^\Omega J_\alpha(r\varpi)^2 r\varpi \, d\varpi \, dr.$$

In particular, if the (ℓ, m) sector is active, then one of its eigenvalues is at least $1/2$, and hence $S_{\ell,\Omega} \geq 1/2$.

It remains to show that this is impossible when ℓ is much larger than Ω. From the power series for J_α, followed by the triangle inequality,

$$\begin{aligned} |J_\alpha(t)| &\leq \frac{(t/2)^\alpha}{\Gamma(\alpha+1)} \sum_{j=0}^\infty \frac{(t^2/4)^j}{j!(\alpha+1)^j} \\ &= \frac{(t/2)^\alpha}{\Gamma(\alpha+1)} \exp\left(\frac{t^2}{4(\alpha+1)}\right). \end{aligned}$$

Here we used $\Gamma(\alpha + j + 1) \geq \Gamma(\alpha + 1)(\alpha + 1)^j$. The elementary Stirling bound $\Gamma(\alpha + 1) \geq (\alpha/e)^\alpha$ then gives

$$|J_\alpha(t)| \leq \left(\frac{et}{2\alpha}\right)^\alpha \exp\left(\frac{t^2}{4(\alpha+1)}\right).$$

Suppose $\alpha \geq 4\Omega$. Throughout the sector-trace integral, $0 \leq t = r\varpi \leq \Omega$, so

$$\frac{et}{2\alpha} \le \frac{e}{8}, \qquad \frac{t^2}{4(\alpha+1)} \le \frac{\alpha}{64}.$$

Thus, with $q = (e/8)e^{1/64} < 1$, we have $|J_\alpha(t)| \le q^\alpha$ uniformly for $0 \le t \le \Omega$. Substituting this into the trace formula yields

$$S_{\ell,\Omega} \le q^{2\alpha} \int_0^1 \int_0^\Omega r\varpi \, d\varpi \, dr = \frac{\Omega^2}{4} q^{2\alpha}.$$

For sufficiently large Ω, the right-hand side is less than $1/2$ whenever $\alpha \ge 4\Omega$. Hence an active angular degree must satisfy $\ell + \nu < 4\Omega$, and in particular $\ell \lesssim \Omega$.

Let L be the largest active degree. Using the formula for $h_{\ell,d}$ and summing the binomial coefficients gives

$$\sum_{\ell=0}^{L} h_{\ell,d} = \binom{L+d-1}{d-1} + \binom{L+d-2}{d-1} \lesssim_d (1+L)^{d-1}.$$

Since $L \lesssim \Omega$, we conclude that

$$H_\Omega \le \sum_{\ell=0}^{L} h_{\ell,d} \lesssim \Omega^{d-1},$$

which proves (Eq. Angular count). $\square$

Normalized features on the input ball

For each retained Slepian mode, define

$$f_\alpha(x) = \sqrt{\frac{V_d}{\lambda_\alpha}}\, \psi_\alpha(x), \qquad f_\alpha^{(A)}(x) = f_\alpha(x/A).$$

Let $F_A(x)$ be the column vector whose entries are the N_Ω retained functions $f_\alpha^{(A)}(x)$.

> **Lemma 3 (normalization and energy bounds).** The retained features have the following properties:
>
> 1. they are orthonormal under μ_A;
> 2. their Fourier transforms are supported in $B_{\Omega/A}$;
> 3. for every coefficient vector $a \in \mathbb{C}^{N_\Omega}$,
>
> $$\int_{\mathbb{R}^d} |a^* F_A(x)|^2 \, dx \lesssim A^d \|a\|^2;$$
>
> 4. uniformly in x,

$$\|F_A(x)\|^2 \lesssim \Omega^d.$$

Proof. On B_1 the Slepian eigenvalue identity gives

$$\int_{B_1} \psi_\alpha(x)\overline{\psi_\beta(x)}\, dx = \lambda_\alpha \delta_{\alpha\beta}.$$

The chosen normalization therefore makes the f_α orthonormal for uniform probability on B_1, and the change of variables $x = Au$ proves orthonormality of $f_\alpha^{(A)}$ under μ_A.

Scaling a function by $x \mapsto x/A$ contracts its Fourier support from B_Ω to $B_{\Omega/A}$, proving the second property. Global orthonormality of the ψ_α and $\lambda_\alpha \geq 1/2$ give

$$\int_{\mathbb{R}^d} |a^* F_A(x)|^2\, dx = V_d A^d \sum_{\alpha \in \mathcal{I}_\Omega} \frac{|a_\alpha|^2}{\lambda_\alpha} \leq 2V_d A^d \|a\|^2.$$

Finally, every $f \in PW_\Omega$ satisfies $f(x) = \langle f, \mathcal{K}_\Omega(x - \cdot)\rangle$. Applying Bessel's inequality to this evaluation formula gives

$$\sum_{\alpha \in \mathcal{I}_\Omega} |\psi_\alpha(x/A)|^2 \leq \mathcal{K}_\Omega(0).$$

Multiplying by $V_d/\lambda_\alpha \leq 2V_d$ yields

$$\|F_A(x)\|^2 \leq 2V_d \mathcal{K}_\Omega(0) \lesssim \Omega^d.$$

□

One shell has only surface-order rank

For a probability law ρ on B_A, define its feature covariance matrix

$$T_\Omega(\rho) = \int_{B_A} F_A(x) F_A(x)^*\, d\rho(x).$$

Orthonormality under the uniform ball gives

$$T_\Omega(\mu_A) = I_{N_\Omega}.$$

Now fix one shell radius r_0. On the sphere $x = r_0\theta$, a separated feature has the form

$$f_{\ell,m,k}^{(A)}(r_0\theta) = a_{\ell,k}(r_0) Y_{\ell,m}(\theta),$$

where $a_{\ell,k}(r_0)$ contains the radial value and the normalization. Angular orthogonality makes $T_\Omega(s_{r_0})$ block diagonal in (ℓ, m). Inside one such block, the entry indexed by (k, k') is

$$a_{\ell,k}(r_0)\overline{a_{\ell,k'}(r_0)},$$

which is an outer product and therefore has rank at most one. By (Eq. Angular count), there are only $H_\Omega \lesssim \Omega^{d-1}$ active angular blocks, so

$$\operatorname{rank} T_\Omega(s_{r_0}) \leq H_\Omega \lesssim \Omega^{d-1}.$$

For a mixture of K shells, subadditivity of rank gives

$$\operatorname{rank} T_\Omega(\rho) \lesssim K\Omega^{d-1}. \tag{2}$$

Choose $\Omega = \kappa_d K$, where the fixed dimensional factor κ_d is large enough. Combining (Eq. Mode count) and (Eq. 2) then yields

$$\operatorname{rank} T_\Omega(\rho) \leq \frac{1}{2} N_\Omega.$$

Let

$$\Delta_\Omega = T_\Omega(\rho) - I_{N_\Omega}.$$

This is precisely the Frobenius-norm case of the Eckart–Young–Mirsky theorem. If $r = \operatorname{rank} T_\Omega(\rho) \leq N_\Omega/2$, then every rank-$r$ approximation to the identity satisfies

$$\begin{aligned} \|\Delta_\Omega\|_F^2 &= \|T_\Omega(\rho) - I_{N_\Omega}\|_F^2 \\ &\geq \sum_{j=r+1}^{N_\Omega} \sigma_j(I_{N_\Omega})^2 = N_\Omega - r \\ &\geq \frac{1}{2} N_\Omega \gtrsim \Omega^d. \end{aligned} \tag{3}$$

Here every singular value $\sigma_j(I_{N_\Omega})$ equals 1. Equivalently, $T_\Omega(\rho)$ has a null space of dimension at least $N_\Omega/2$, and Δ_Ω acts as $-I$ on that null space. This is the vector analogue of the scalar Toeplitz rank defect.

Transfer through Gaussian smoothing

The matrix defect concerns expectations of functions of the input. We now build a test function that reads the same defect from the output density.

For a Hermitian matrix M, define the quadratic feature

$$u_M(x) = F_A(x)^* M F_A(x).$$

Lemma 3 implies

$$\|u_M\|_2^2 \le \left(\sup_x \|F_A(x)\|^2\right) \int_{\mathbb{R}^d} \|MF_A(x)\|^2\,dx \lesssim A^d \Omega^d \|M\|_F^2. \tag{4}$$

Every entry of F_A is bandlimited to $B_{\Omega/A}$. Products of two such entries are therefore bandlimited to $B_{2\Omega/A}$, so

$$\operatorname{supp} \hat{u}_M \subseteq B_{2\Omega/A}.$$

Convolution with the Gaussian law ϕ_d multiplies the Fourier transform by $e^{-\|\xi\|^2/2}$. Define h_M by reversing this multiplier on the finite frequency ball:

$$\hat{h}_M(\xi) = e^{\|\xi\|^2/2} \hat{u}_M(\xi).$$

Because $\hat{u}_M$ vanishes outside $B_{2\Omega/A}$, Plancherel and (Eq. 4) give

$$\|h_M\|_2^2 \lesssim A^d \Omega^d \exp\left(\frac{4\Omega^2}{A^2}\right) \|M\|_F^2. \tag{5}$$

By construction, $h_M * \phi_d = u_M$. Using the symmetry of the Gaussian density and Fubini's theorem,

$$\int_{\mathbb{R}^d} h_M(y)\big(p_\rho(y) - p_{\mu_A}(y)\big)\,dy = \int_{B_A} u_M(x)\,d(\rho - \mu_A)(x).$$

The right-hand side is the Frobenius inner product $\langle M, \Delta_\Omega \rangle_F = \operatorname{tr}(M^* \Delta_\Omega)$. Take $M = \Delta_\Omega$. Cauchy–Schwarz, (Eq. 3), and (Eq. 5) yield

$$\begin{aligned}
\|p_\rho - p_{\mu_A}\|_2^2 &\ge \frac{\|\Delta_\Omega\|_F^4}{\|h_{\Delta_\Omega}\|_2^2} \\
&\gtrsim \frac{\|\Delta_\Omega\|_F^2}{A^d \Omega^d} \exp\left(-\frac{4\Omega^2}{A^2}\right) \\
&\gtrsim \frac{1}{A^d} \exp\left(-c_d \frac{K^2}{A^2}\right).
\end{aligned}$$

The last line uses $\Omega = \kappa_d K$. Multiplying by A^d proves (Eq. Approx. lower bound).

Step 1: Jeffreys is near the CAID

Let $\mathsf{C}_d(A)$ be the channel capacity. For any input law ρ on B_A, the capacity-achieving output satisfies the stability identity

$$I_\rho(X;Y) + D(p_\rho \| p_{\rho_A^*}) = \int_{B_A} D(\phi_d(\cdot - x) \| p_{\rho_A^*})\, d\rho(x) \le \mathsf{C}_d(A).$$

The final inequality is the KKT inequality for the CAID. Taking $\rho = \mu_A$ gives

$$D(p_{\mu_A} \| p_{\rho_A^*}) \le \mathsf{C}_d(A) - I_{\mu_A}(X;Y).$$

Monotonicity of entropy under independent Gaussian convolution gives

$$I_{\mu_A}(X;Y) \ge \log(V_d A^d) - \frac{d}{2}\log(2\pi e).$$

The vector capacity upper bound of Thangaraj, Kramer, and Böcherer has the high-amplitude form

$$\mathsf{C}_d(A) \le \log\left(\frac{V_d A^d}{(2\pi e)^{d/2}} + O_d(A^{d-1})\right).$$

Subtracting the mutual-information lower bound gives

$$D(p_{\mu_A} \| p_{\rho_A^*}) \lesssim \log\left(1 + \frac{1}{A}\right) \lesssim \frac{1}{A},$$

which proves (Eq. Jeffreys–CAID).

Step 3: bridging Steps 1 and 2

We next show that the reverse KL divergence in Step 1 controls the L^2 distance in Step 2. The key fact is that both output densities have height of order at most A^{-d}.

For the uniform input,

$$p_{\mu_A}(y) = \frac{1}{V_d A^d} \int_{B_A} \phi_d(y - x)\, dx \le \frac{1}{V_d A^d}.$$

The optimal output requires a little more work. In the argument below, inequalities between symmetric matrices are understood in the quadratic-form sense.

Lemma 4 (height of the optimal output). The CAID output satisfies

$$\|p_{\rho_A^*}\|_\infty \le \frac{e^d}{V_d A^d}.$$

Proof. For a Gaussian mixture, differentiation under the integral gives the score and Hessian identities

$$\nabla \log p_{\rho_A^*}(y) = \mathbb{E}[X^* \mid Y = y] - y,$$

and

$$\nabla^2 \log p_{\rho_A^*}(y) = \mathrm{Cov}(X^* \mid Y = y) - I_d.$$

Let y_0 maximize $p_{\rho_A^*}$ and write $M_A = p_{\rho_A^*}(y_0)$. At y_0, the score vanishes and the Hessian is negative semidefinite, hence

$$\mathbb{E}[X^* \mid Y = y_0] = y_0, \qquad \mathrm{Cov}(X^* \mid Y = y_0) \preceq I_d.$$

The conditional mean-square distance from y_0 is therefore at most d, so the support of ρ_A^* contains a point x_0 satisfying $\|x_0 - y_0\|^2 \leq d$. The Hessian identity also gives $\nabla^2 \log p_{\rho_A^*} \succeq -I_d$ everywhere. Integrating this lower curvature bound away from the maximizer yields

$$\log p_{\rho_A^*}(y) \geq \log M_A - \frac{1}{2}\|y - y_0\|^2.$$

The KKT equality at the support point x_0 is

$$\mathbb{E} \log p_{\rho_A^*}(x_0 + Z) = -\mathsf{C}_d(A) - h(Z).$$

Using the preceding quadratic lower bound and $\mathbb{E}\|Z\|^2 = d$ gives

$$-\mathsf{C}_d(A) - h(Z) \geq \log M_A - d.$$

Finally,

$$\mathsf{C}_d(A) \geq I_{\mu_A}(X; Y) \geq \log(V_d A^d) - h(Z),$$

and the claimed bound on M_A follows. □

We now use a general comparison between KL divergence, Hellinger distance, and L^2 distance. For densities p and q, define

$$H^2(p, q) = \int_{\mathbb{R}^d} (\sqrt{p} - \sqrt{q})^2.$$

If both densities are bounded by B, then

$$\|p - q\|_2^2 = \int (\sqrt{p} - \sqrt{q})^2 (\sqrt{p} + \sqrt{q})^2 \leq 4BH^2(p, q).$$

On the other hand, Jensen's inequality gives

$$
\begin{aligned}
D(q\|p) &= -2\int q\log\sqrt{p/q} \\
&\geq -2\log\int\sqrt{pq} \\
&\geq 2\left(1-\int\sqrt{pq}\right) = H^2(p,q).
\end{aligned}
$$

Apply these inequalities with $q = p_{\mu_A}$, $p = p_{\rho_A^*}$, and $B = e^d/(V_d A^d)$. We obtain

$$
D(p_{\mu_A}\|p_{\rho_A^*}) \geq H^2(p_{\rho_A^*}, p_{\mu_A}) \gtrsim A^d\|p_{\rho_A^*} - p_{\mu_A}\|_2^2,
$$

which proves (Eq. Bridge).

Remarks on dimension-dependent constants

The notation $\lesssim_d$ and $\gtrsim_d$ hides multiplicative constants depending only on the fixed dimension. They enter through the volume V_d, the vector capacity bound, the multiplicities of spherical harmonics, the Bessel estimates, and the choice of the proportionality factor in $\Omega \asymp K$.

The exponent constant c_d is displayed because changing it changes the scale of K_A^2/A^2 and cannot be absorbed into an outside prefactor. No attempt is made here to optimize this constant or to obtain a bound uniform in growing dimension. The statement is asymptotic in A for each fixed d.

Conclusion

The proof can be summarized by the chain

$$
\boxed{\text{few shells} \Longrightarrow \text{low angular rank} \Longrightarrow \text{spectral defect} \Longrightarrow \text{output separation.}}
$$

The Slepian modes provide volume-order many probes of the ball, while a single spherical shell can activate only surface-order many angular channels. Choosing the spectral radius proportional to the number of shells creates the missing directions that survive Gaussian smoothing strongly enough to contradict near-optimality unless K_A has the asserted size.

Suggested citation

H. Wang, "Technical proof of the extension to higher-dimensional Gaussian channels," Technical Report HW-TR-2026-01, Yale University, 2026. PDF · BibTeX

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